

SUPER-STORE SEALS ANALYSIS PROJECT

[END-TO-END SQL AND
PYTHON PROJECT]

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1. Introduction

In today's data-driven world, organizations generate large volumes of data from various sources. However, raw data alone has little value unless it is properly organized, analyzed, and interpreted. Data analytics helps transform raw data into meaningful insights that support informed decision-making and improve business performance.

This project demonstrates an end-to-end data analysis process using SQLite (SQL) and Python. SQLite is used for storing, cleaning, and querying the dataset, while Python is used for data processing, exploratory data analysis (EDA), and data visualization. Together, these tools enable efficient analysis of real-world datasets. The project includes database creation, data cleaning, SQL-based analysis, Python integration, and visualization to identify patterns, trends, and key insights. Overall, it showcases how SQL and Python can be combined to convert raw data into valuable business insights and support data-driven decision-making.

2. Project Objective

The primary objective of this project is to perform an end-to-end data analysis using SQLite (SQL) and Python to extract meaningful insights from a real-world dataset. The project aims to demonstrate the complete data analytics workflow, including data storage, cleaning, querying, analysis, and visualization.

The specific objectives of this project are:

- To import and organize the dataset in an SQLite database.
 - To clean and preprocess the data for accurate analysis.
 - To use SQL queries to explore the dataset and extract meaningful information.
 - To integrate SQLite with Python for advanced data analysis.
 - To perform exploratory data analysis (EDA) using Python libraries.
 - To create clear and informative visualizations that highlight trends and patterns.
 - To generate business insights and provide data-driven recommendations based on the analysis.
 - To strengthen practical skills in SQL, Python, database management, and data visualization through a real-world analytics project.
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3. Problem Statement

The project aims to address the following challenges:

- Large volumes of raw data are often difficult to understand and interpret.
 - Unclean or inconsistent data can lead to inaccurate analysis and unreliable results.
 - Extracting meaningful insights from raw data requires proper data management and analysis techniques.
 - Identifying trends, patterns, and key performance indicators manually is time-consuming and inefficient.
 - Businesses need clear visualizations and data-driven insights to support effective decision-making.
 - There is a need to combine SQL and Python to efficiently clean, analyze, and visualize the dataset.
 - The project focuses on transforming raw data into meaningful insights that help solve real-world business problems.
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4. Dataset Description

The dataset used in this project serves as the primary source for analysis. It contains structured information that enables meaningful exploration and insight generation.

Dataset Details:

- Dataset Name: (Super-Store)
- Source: (Kaggle)
- Number of Rows: (9994)
- Number of Columns: (12)
- File Format: CSV / Excel
- Database Used: SQLite

Dataset Features:

- Contains both numerical and categorical data.
- Includes multiple attributes relevant to the selected domain.
- Suitable for SQL querying, exploratory data analysis (EDA), and visualization.
- Used to identify trends, patterns, and key performance indicators (KPIs).
- Cleaned and preprocessed by handling missing values, duplicates, and inconsistent data before analysis.

This dataset provides the foundation for performing SQL-based analysis, Python integration, and data visualization to generate meaningful business insights and support data-driven decision-making.

5. Technology Stack

The following technologies, tools, and libraries were used to develop and analyze this project:

Technology / Tool	Purpose
SQLite	Store and manage the dataset in a relational database.
SQL	Perform data cleaning, querying, filtering, and analysis.
SQLiteViz	Execute SQL queries and visualize database tables.
Python	Perform data analysis and integrate with the SQLite database.
Pandas	Load, clean, manipulate, and analyze data.
NumPy	Perform numerical operations and calculations.
Matplotlib	Create static charts and graphs.
Plotly	Develop interactive and visually appealing data visualizations.
Google Colab / Jupyter Notebook	Write, execute, and document Python code.

6. Project Workflow

The project follows a structured workflow to ensure systematic data analysis and accurate results. The major steps involved are:

- **Dataset Collection** – Obtain the dataset from a reliable source.
 - **Data Import** – Import the dataset into an SQLite database.
 - **Database Creation** – Create the database schema and organize the data.
 - **Data Cleaning** – Remove duplicates, handle missing values, and correct inconsistencies.
 - **SQL Analysis** – Execute SQL queries to explore the data and extract meaningful information.
 - **Python Integration** – Connect the SQLite database with Python for further analysis.
 - **Exploratory Data Analysis (EDA)** – Analyze the data using Python libraries such as Pandas and NumPy.
 - **Data Visualization** – Create charts and graphs using Matplotlib and Plotly to identify trends and patterns.
 - **Business Insights** – Interpret the analysis results to uncover key findings.
 - **Recommendations & Conclusion** – Provide actionable recommendations and summarize the overall findings of the project.
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7. SQL Database Setup

7.1 Import Dataset

The dataset was imported into an SQLite database to enable efficient storage, management, and querying of the data. Importing the dataset into a relational database provides a structured environment for performing data cleaning and analysis using SQL.

7.2 Data Cleaning

Before performing the analysis, the dataset was cleaned to improve data quality and reliability. This process included handling missing values, removing duplicate records, correcting inconsistent data, and validating data types to ensure accurate analytical results.

```
-- SQL DATA CLEANING

SELECT * FROM "Superstore"
limit 10;

SELECT COUNT(*) total_rows
FROM Superstore;

SELECT *
FROM Superstore
WHERE Sales IS NULL;

SELECT
    "Order ID",
    COUNT(*)
FROM Superstore
GROUP BY "Order ID"
HAVING COUNT(*) > 1;

SELECT *
FROM Superstore
WHERE Profit<0;

SELECT DISTINCT Discount
FROM superstore
ORDER BY Discount;
```

7.3 SQL Queries

SQL queries were written to explore the dataset, retrieve relevant information, calculate key metrics, and answer important analytical questions. Various SQL operations such as filtering, sorting, grouping, aggregation, and conditional statements were used to uncover meaningful insights from the data.

```
-- NOW SQL BUSINESS ANALYSIS
-- Basic KPIs

-- TOTAL SALES
SELECT
    ROUND(SUM(Sales), 2) AS Sales
FROM Superstore;

-- TOTAL PROFIT
SELECT
    ROUND(SUM(Profit), 2) AS Profit
FROM Superstore;

-- TOTAL ORDERS
SELECT
    COUNT(DISTINCT "Order ID") AS Total_Orders
FROM Superstore;

-- TOTAL CUSTOMERS
SELECT
    COUNT(DISTINCT "Customer ID") AS Total_Customers
FROM Superstore;

-- TOTAL PRODUCTS
SELECT
    COUNT(DISTINCT "Product ID") AS Total_Products
FROM Superstore;

-- Geographic Analysis
-- Sales by Region
SELECT
    Region,
    ROUND(SUM(Sales),2) AS Total_Sales
FROM Superstore
GROUP BY Region
ORDER BY Total_Sales DESC;

-- Profit by Region
SELECT
    Region,
    ROUND(SUM(Profit),2) AS Total_Profit
FROM Superstore
GROUP BY Region
ORDER BY Total_Profit DESC;

-- Top 10 States by Sales
SELECT
    State,
    ROUND(SUM(Sales),2) AS Total_Sales
FROM Superstore
GROUP BY State
ORDER BY Total_Sales DESC
LIMIT 10;
```

8. Python Integration

8.1 Connecting SQLite with Python

After completing the SQL analysis, the SQLite database was connected to Python using the built-in `sqlite3` library. This integration enabled SQL query results to be loaded directly into Pandas DataFrames for further analysis. Python was then used to perform exploratory data analysis (EDA), process the retrieved data, and generate meaningful visualizations.

```
plt.style.use("ggplot")
```

```
conn = sqlite3.connect("/content/sample_data/database.sqlite")
```

8.2 Libraries Used

Several Python libraries were used throughout the project to perform data analysis and visualization. Each library served a specific purpose in the analytics workflow.

Library	Purpose
sqlite3	Connect Python with the SQLite database and execute SQL queries.
Pandas	Load, clean, manipulate, and analyze the dataset.
NumPy	Perform numerical operations and mathematical calculations.
Matplotlib	Create static charts and graphs for data visualization.
Plotly	Develop interactive and visually appealing visualizations.

```
import sqlite3
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

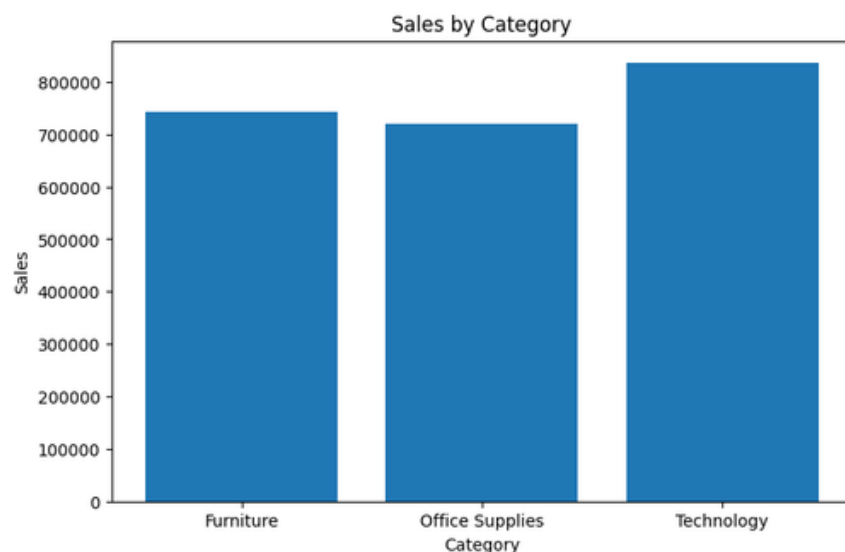
9. Data Visualization

Data visualization is an essential part of data analysis, as it helps transform complex datasets into clear and meaningful graphical representations. Visualizations make it easier to identify trends, patterns, relationships, and anomalies that may not be immediately apparent from raw data or tables.

In this project, various charts and graphs were created using Matplotlib and Plotly to effectively present the results of the analysis. Each visualization focuses on a specific aspect of the dataset and provides valuable insights to support data-driven decision-making.

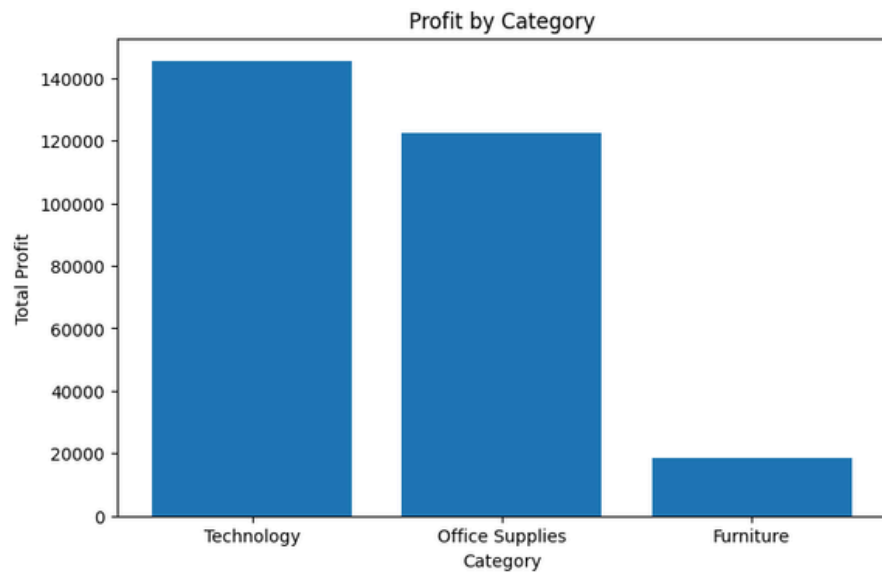
9.1 Sales by Category

This bar chart compares the total sales generated by each product category.



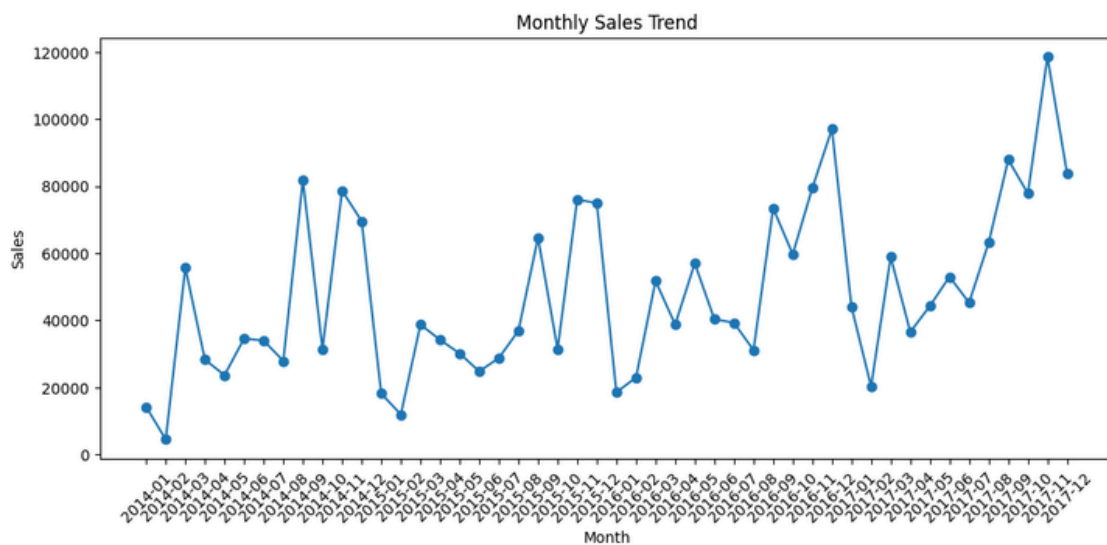
9.2 Profit by Category

This bar chart illustrates the total profit earned from each product category.



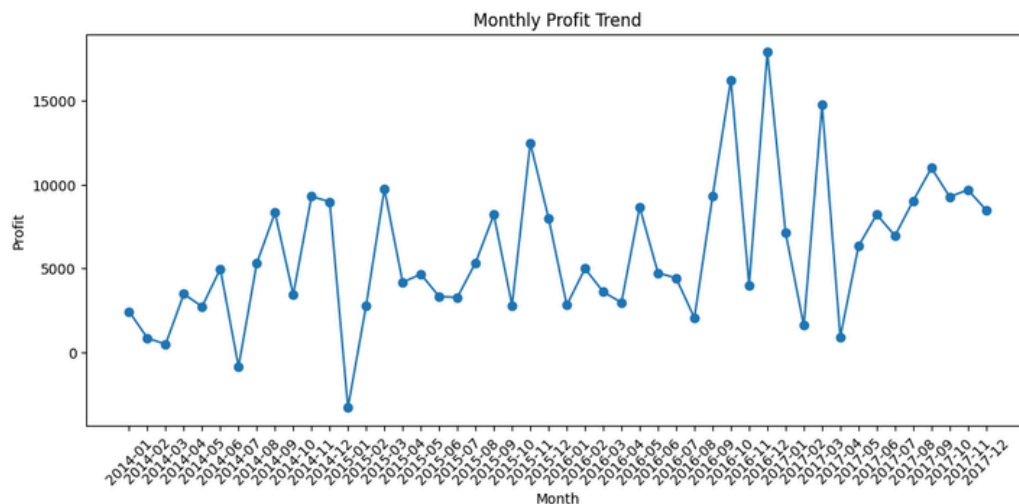
9.3 Monthly Sales Trend

This line chart shows the variation in monthly sales over the analysis period.



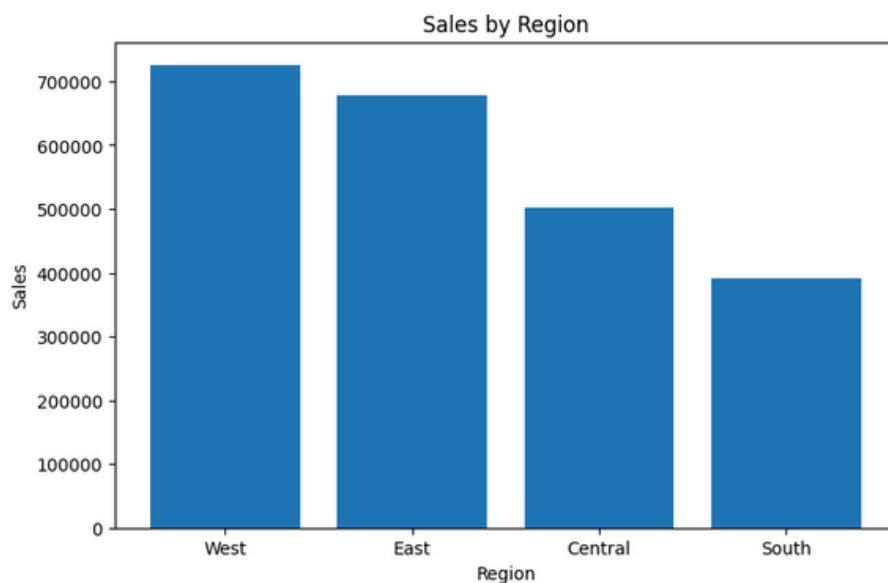
9.4 Monthly Profit Trend

This line chart represents the monthly profit trend and highlights changes in profitability over time.



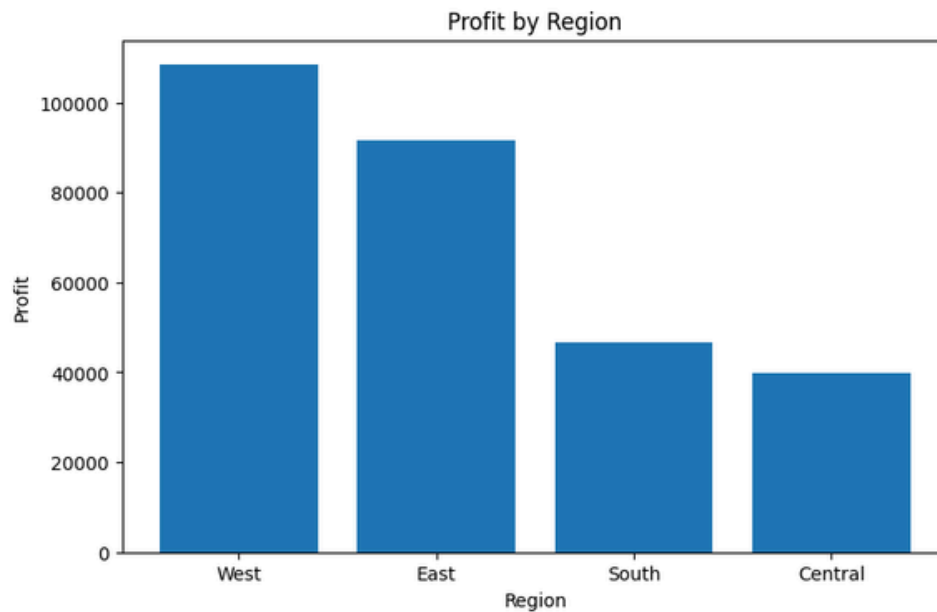
9.5 Sales by Region

This bar chart compares total sales across different geographical regions.



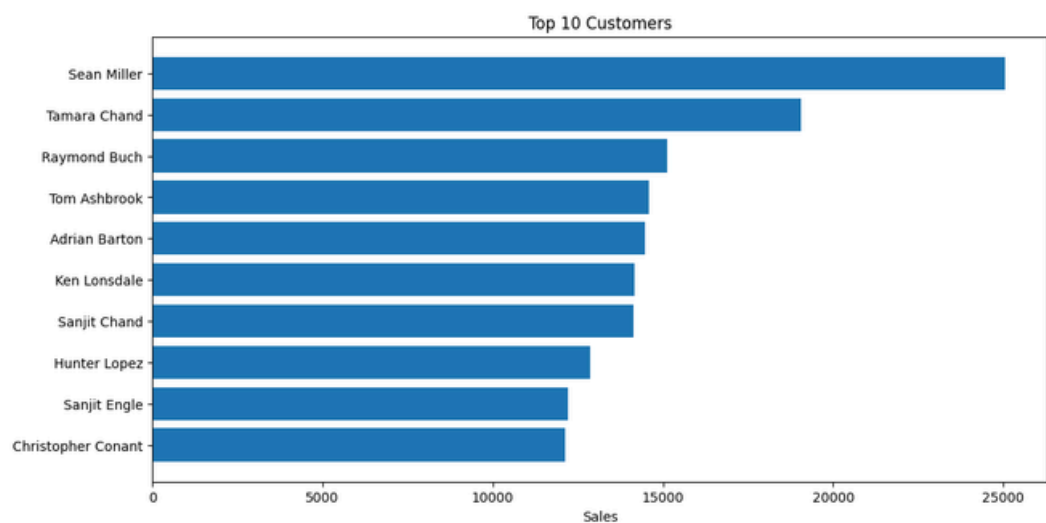
9.6 Profit by Region

This bar chart displays the total profit generated by each region.



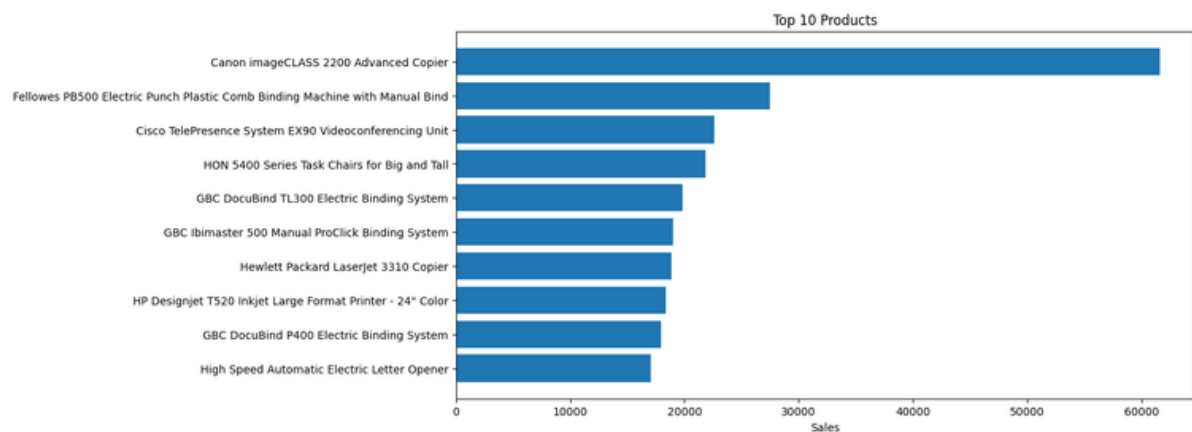
9.7 Top 10 Customers

This horizontal bar chart identifies the top 10 customers based on total sales.



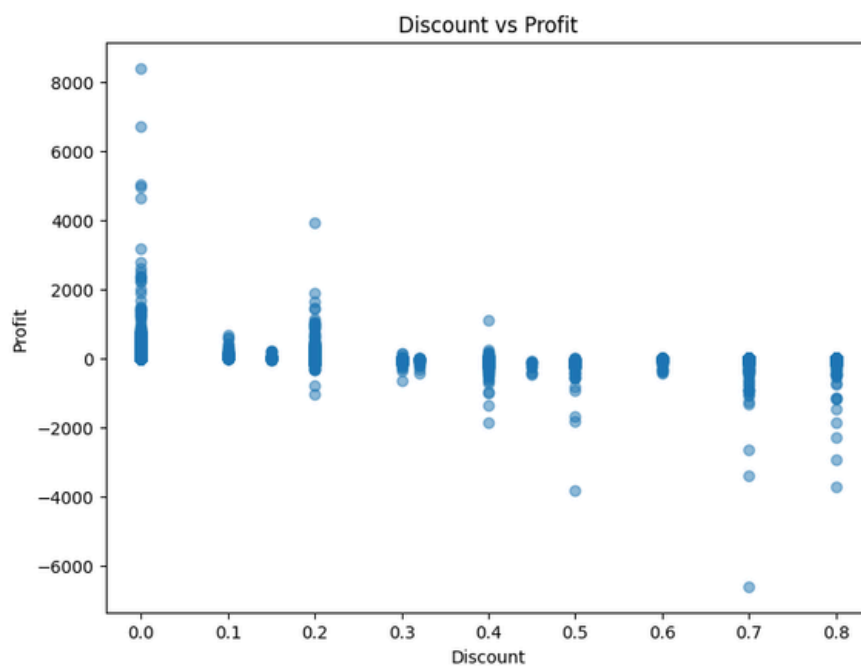
9.8 Top 10 Products

This horizontal bar chart presents the top 10 products with the highest sales.



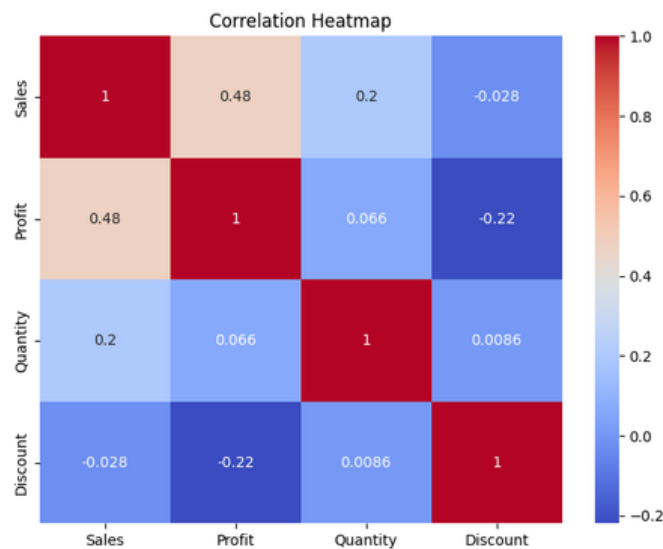
9.9 Discount vs Profit

This scatter plot illustrates the relationship between discount offered and the resulting profit.



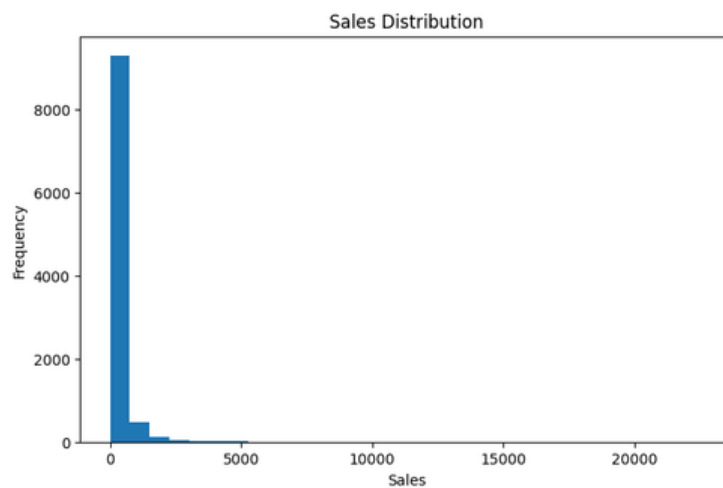
9.10 Correlation Heatmap

This heatmap visualizes the correlation between numerical variables such as Sales, Profit, Quantity, and Discount.



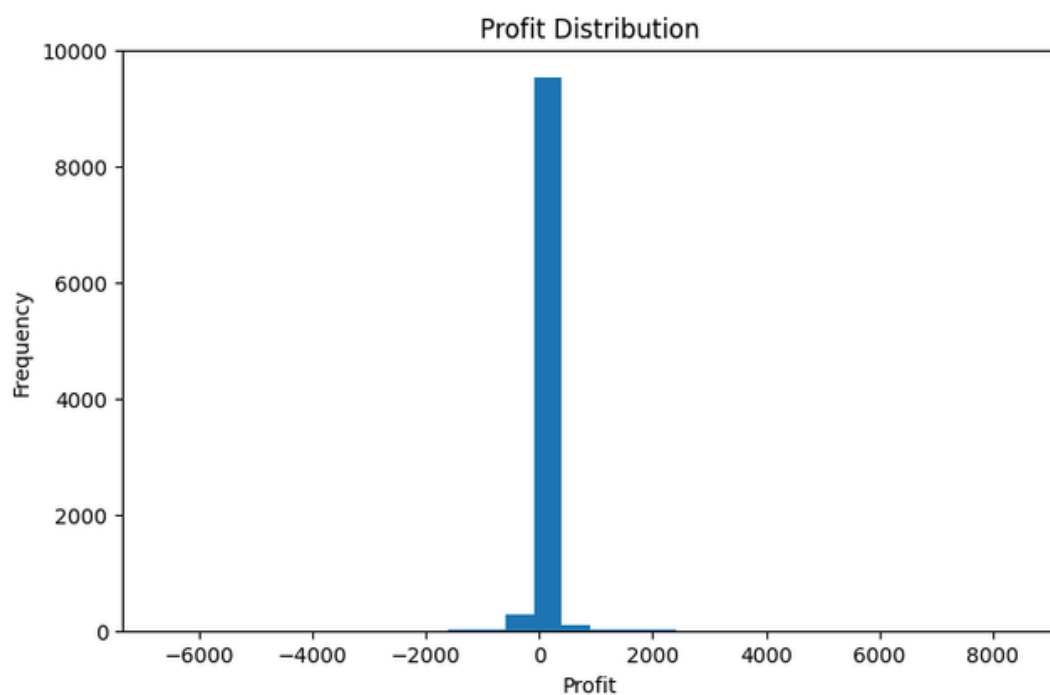
9.11 Sales Distribution

This histogram shows the frequency distribution of sales values across the dataset.



9.12 Profit Distribution

This histogram illustrates the distribution of profit values, highlighting the spread of profits and losses.



10. Conclusion

This project successfully demonstrated an end-to-end data analysis process using SQLite (SQL) and Python. The dataset was imported, organized, and cleaned in an SQLite database before being analyzed through SQL queries to extract meaningful information. The processed data was then integrated with Python for exploratory data analysis (EDA) and visualization using libraries such as Pandas, NumPy, Matplotlib, and Plotly.

The analysis revealed valuable trends, patterns, and relationships within the dataset, which were presented through clear and informative visualizations. These insights can support better business decision-making and improve overall understanding of the data.

Overall, this project highlights the effective use of SQL and Python in managing, analyzing, and visualizing real-world data. It also demonstrates practical skills in database management, data analysis, and data visualization, making it a strong portfolio project for showcasing end-to-end data analytics capabilities.
